Lecture 20-22

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1 Functions

1.1 Definition

Let X and Y be two sets. What is a function from X to Y? A **function** f from X to Y is a relation $f \subseteq X \times Y$ such that for every $x \in X$, there exists exactly one $y \in Y$ such that $(x, y) \in f$.

1.2 Notation

$$f : X \to Y$$

$$x \mapsto y \quad \text{or} \quad y = f(x) \iff (x, y) \in f$$

- $x \in X$ is an **argument**.
- $y = f(x)$ is a **value**.
- The function maps X **into** Y.
- Synonyms: map, mapping, correspondence, transformation.

1.3 Graph of a Function

The set $\{(x, f(x)) : x \in X\}$ is called the **graph** of f.

1.4 Domain and Range

For $f : X \to Y$:

- The set X is called the **domain** of f.
- The set Y is called the **codomain** (or target set) of f.
- The set of actual values $\{f(x) : x \in X\}$ is called the **range** or **image** of f.

1.5 Examples

1. Function from $\mathbb{Z}$ to $\mathbb{Z}$:

   - $f = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : 3x + 2y = 0\}$ If $x = 1$, then $3(1) + 2y = 0 \implies 2y = -3$, which has no integer solution. Thus, this relation is **not** a function on $\mathbb{Z}$.
   - $g = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : y = 0\}$. This is a function $g(x) = 0$.

2. Defining Functions:

   - $f_1 : \mathbb{R} \setminus \{2\} \to \mathbb{R}$ defined by $f_1(x) = \frac{x^2-4}{x-2}$.
   - $f_2 : \mathbb{R} \setminus \{2\} \to \mathbb{R}$ defined by $f_2(x) = x + 2$.
   - Are f_1 and f_2 equal? **YES!** Because their domains are the same and for all x in the domain, $\frac{x^2-4}{x-2} = x + 2$.
   - Note: $g : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = x + 2$ is **not** equal to f_1 because the domains differ (f_1 is not defined at $x = 2$).

3. Function on pairs:

   - $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ defined by $f(m, n) = m - n$.

2 Properties of Functions

Let $f : X \to Y$.

2.1 Equality of Functions

Two functions $f : X_1 \to Y_1$ and $g : X_2 \to Y_2$ are **equal** if:

1. $X_1 = X_2$ (Same Domain).
2. $Y_1 = Y_2$ (Same Codomain - sometimes omitted depending on context).
3. $f(x) = g(x)$ for all $x \in X_1$.

2.2 Injectivity (One-to-One)

A function f is **injective** (or one-to-one) if:

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$

Equivalently:

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

Examples:

- $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ is **not** injective ($f(1) = f(-1)$).
- $f : \mathbb{R} \to \mathbb{R}, f(x) = x^3$ is injective.
- $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}, f(x) = \frac{1}{x} + 1$ is injective.

2.3 Surjectivity (Onto)

A function f is **surjective** (or onto) if:

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y$$

This means the range of f is equal to the codomain Y ($f(X) = Y$).

Examples:

- $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ is **not** surjective (negative numbers are not in the image).
- $f : \mathbb{R} \to \mathbb{R}_{\geq 0}, f(x) = x^2$ **is** surjective.
- $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}, f(x) = \frac{1}{x} + 1$ is **not** surjective (cannot take value 1).

2.4 Bijectivity

A function f is **bijective** if it is both injective and surjective.

Example:

- $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ is bijective.

2.5 Complex Example

Let $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ defined by $f(m, n) = (m + n, m + 2n)$.

Injectivity Check: Suppose $f(m, n) = f(a, b)$.

$$(m + n, m + 2n) = (a + b, a + 2b)$$

$$\implies \begin{cases} m + n = a + b \\ m + 2n = a + 2b \end{cases}$$

Subtracting the first eq from the second: $n = b$. Substituting back: $m + b = a + b \implies m = a$. Thus $(m, n) = (a, b)$. f is injective.

Surjectivity Check: Let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$. Can we find (m, n) such that $f(m, n) = (a, b)$?

$$\begin{cases} m + n = a \\ m + 2n = b \end{cases}$$

Subtracting: $n = b - a$. Then $m = a - n = a - (b - a) = 2a - b$. Since $a, b \in \mathbb{Z}$, m and n are integers. Thus f is surjective. **Conclusion:** f is bijective.

3 Composition of Functions

3.1 Definition

Let $f : X \to Y$ and $g : Y \to Z$. The **composition** of g and f is the function $g \circ f : X \to Z$ defined by:

$$(g \circ f)(x) = g(f(x))$$

Note: $g \circ f$ makes sense when codomain of f matches domain of g. $f \circ g$ usually makes no sense unless $X = Z$. Even if defined, $f \circ g \neq g \circ f$ generally.

3.2 Associativity

Composition is associative. Let $f : A \to B, g : B \to C, h : C \to D$. Then $(h \circ g) \circ f = h \circ (g \circ f)$. **Proof:** For any $x \in A$:

$$((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x)))$$

$$(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x)))$$

The domains and codomains match.

3.3 Properties under Composition

Let $f : X \to Y$ and $g : Y \to Z$.

1. If f and g are injective $\implies g \circ f$ is injective. **Proof:** $(g \circ f)(x_1) = (g \circ f)(x_2) \implies g(f(x_1)) = g(f(x_2))$. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$.

2. If f and g are surjective $\implies g \circ f$ is surjective. **Proof:** For any $z \in Z$, since g is surjective, $\exists y \in Y$ such that $g(y) = z$. Since f is surjective, $\exists x \in X$ such that $f(x) = y$. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$.

4 Inverse Functions

4.1 Identity Function

The **identity function** on X is $id_X : X \to X$ defined by $id_X(x) = x$.

4.2 Inverse Relation

Let $R \subseteq X \times Y$ be a relation. The **inverse relation** $R^{-1} \subseteq Y \times X$ is:

$$R^{-1} = \{(y, x) \in Y \times X : (x, y) \in R\}$$

4.3 Inverse Function Definition

Let $f : X \to Y$ be a **bijection**. The **inverse function** $f^{-1} : Y \to X$ is the inverse relation of f.

$$f^{-1}(y) = x \iff f(x) = y$$

Properties:

- f^{-1} is a function only if f is bijective.
- $f^{-1} \circ f = id_X$
- $f \circ f^{-1} = id_Y$
- Note: $f^{-1} \neq \frac{1}{f}$.

4.4 Calculation of Inverses

1. $f : \mathbb{R} \to \mathbb{R}, f(x) = x^3 - 1$. Let $y = x^3 - 1 \iff x^3 = y + 1 \iff x = \sqrt[3]{y+1}$. Thus $f^{-1}(y) = \sqrt[3]{y+1}$.
2. $f : \mathbb{R}^2 \to \mathbb{R}^2, f(x, y) = ((x^2 + 1)y, x^3)$. Let $(a, b) = ((x^2 + 1)y, x^3)$. From 2nd component: $x^3 = b \implies x = b^{1/3}$. From 1st component: $(x^2 + 1)y = a \implies (b^{2/3} + 1)y = a \implies y = \frac{a}{b^{2/3}+1}$. Thus $f^{-1}(a, b) = (b^{1/3}, \frac{a}{b^{2/3}+1})$.
3. $f : \mathbb{Z}^2 \to \mathbb{Z}^2, f(m, n) = (5m + 4n, 4m + 3n)$. Solving the system for m, n yields: $f^{-1}(a, b) = (4b - 3a, 4a - 5b)$. (Checking determinant $5(3) - 4(4) = -1$, so inverse exists in integers).

5 Image and Preimage

Let $f : X \to Y$, $A \subseteq X$, $B \subseteq Y$.

5.1 Definitions

- **Image of A:** $f(A) = \{y \in Y : \exists a \in A, f(a) = y\} = \{f(a) : a \in A\}$.
- **Preimage of B:** $f^{-1}(B) = \{x \in X : f(x) \in B\}$. (Note: $f^{-1}(B)$ makes sense even if f has no inverse function).

5.2 Example

$f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$.

- $f(\{1, 2\}) = \{1, 4\}$.
- $f^{-1}(\{1, 4\}) = \{-1, 1, -2, 2\}$.
- $f^{-1}(f(A)) \neq A$ generally (e.g., $f^{-1}(f(\{1\})) = \{-1, 1\} \neq \{1\}$).
- $f^{-1}((1, 9)) = (-3, -1) \cup (1, 3)$.
- $f((-\infty, 1)) = [0, \infty)$ (assuming typo in notes, usually range is non-negative).

5.3 Set Theoretic Properties

Let $A, B \subseteq X$ and $C, D \subseteq Y$.

1. $f(A \cap B) \subseteq f(A) \cap f(B)$ (Equality does not hold generally).
2. $f(A \cup B) = f(A) \cup f(B)$.
3. $A \subseteq f^{-1}(f(A))$.
4. $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.
5. $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.
6. $f(f^{-1}(C)) \subseteq C$.

Proof of (5): $x \in f^{-1}(C \cup D) \iff f(x) \in C \cup D \iff f(x) \in C \vee f(x) \in D \iff x \in f^{-1}(C) \vee x \in f^{-1}(D) \iff x \in f^{-1}(C) \cup f^{-1}(D)$.

6 Cardinality and Functions on Finite Sets

Let X, Y be finite sets.

1. If $f : X \to Y$ is **injective**, then $|X| \leq |Y|$. **Proof:** $f(X) \subseteq Y$ has $|X|$ distinct elements. Thus $|X| = |f(X)| \leq |Y|$.

2. If $f : X \to Y$ is **surjective**, then $|X| \geq |Y|$. **Proof:** $|f(X)| = |Y|$. Since the image can't have more elements than the domain, $|Y| \leq |X|$.

3. If $f : X \to Y$ is **bijective**, then $|X| = |Y|$.